engineering 4 us all

Student Handout

Name: ______________________ **Date:** ______________ **Class:** ______________

MyDesign Portfolio

Element I Initial Template

1. The prototype must support an organizational method

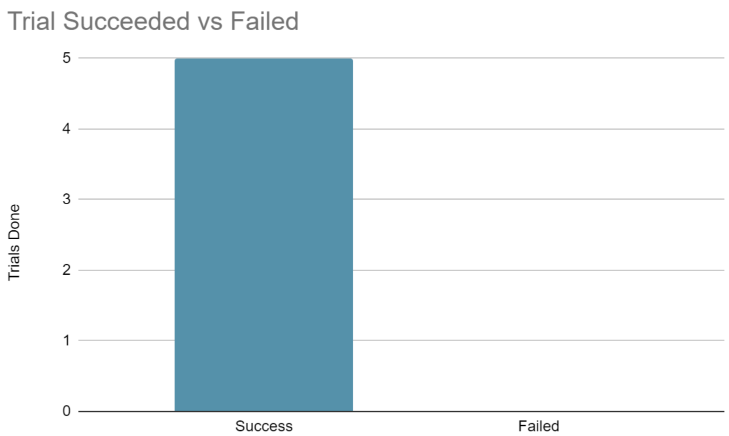
   High priority

   Locate one roll of a specific type of filament, and continue to do this step until you have located one of each type or color of filament

   The IV is placement of the specific colored roll, DV is time taken to find the rolls

   The two outcomes: Color roll found or not found in the time given

   Data: All 7 trials were a success

   As the chart shows 100% of the trials conducted were successful. Based on our testing the shelves allow for multiple organizational methods, and maintain an ease of location of various colors of rolls.

2. The prototype must hold up to 200 spools of filament

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High priority

Assemble all 200 rolls of filament

Stack the rolls filling the top shelf first

Once the top shelf falls put the remaining rolls on the horizontal pegs

If any rolls remain count up the amount leftover

The IV is the amount of rolls on the pegs, the DV is if the shelves hold or not.

Data: 1 trial: The shelf held when all rolls of the filament was placed on it.

Based on our testing data, we found that the self will hold the needed filament.

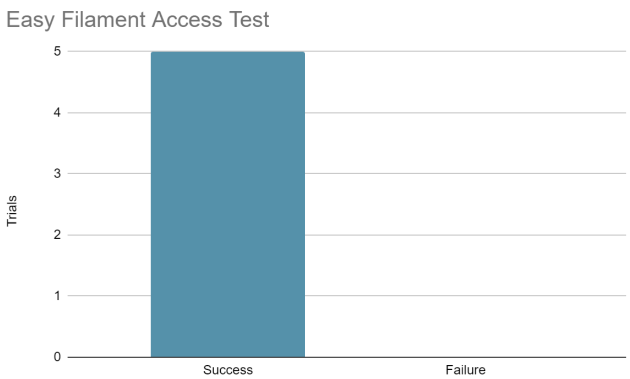
3. The prototype must allow easy access to all the types of filaments it holds

High priority

Process

First, grab one roll of a specific type of filament. Continue to do this step until you have collected one of each type or color of filament. If any variety of roll remains count the number of different varieties left after taking one roll per peg
The IV is the placement of the rolls, the DV is whether a specific variety of filament is accessible after removing one roll of filament prior for all colors.
There were 5 trials, and 5 successes with 0 failures.

Chart



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Summary, Data Meaning and Implication

Based on the testing data – and as is visible in the chart – 100% of the trials conducted were successful, meaning our prototype was successful for this parameter and the shelves allow for easy access to all the types of filaments it holds. Due to the complete success of the prototype (0 failures), it seems that this design is very strong at allowing easy access to filaments, and we should stick with it.